UNIVERSITY OF BUEA

FACULTY OF ENGINEERING AND TECHNOLOGY (FET)

PO BOX 63

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TASK 3: REQUIREMENT ANALYSIS REPORT

COURSE TITLE: Internet Programming & Mobile Programming
COURSE INSTRUCTOR: **Dr. Nkemeni Valery**

SUBMITTED BY
GROUP 3 MEMBERS

NAME	MAT
KONEH CATHERINE MBEH	FE23A076
MOFA DIVINE ESIMO	FE23A095
TUBE NDIANG MARY SHARON	FE23A051
AGBORNJA BESONG AKPAK DIVINE	FE23A006
OBIOMA OBI FODJO CAREL	FE23A133

DEPARTMENT OF COMPUTER ENGINEERING

Table of Contents

1. Introduction 4
2. Review and Analysis of Gathered Requirements 4
2.1 Completeness 4
2.2 Clarity 5
2.3 Technical Feasibility 5
2.4 Dependency Relationships 6
3. Inconsistencies, Ambiguities and Missing Information 6
Identifying Inconsistencies 7
Identifying Ambiguities 7
Identifying Missing Information 8
Importance of Resolving These Issues 9
4. Requirement Prioritisation 9
4.1 Must Have 9
4.2 Should Have 10
4.3 Could Have 10
4.4 Won't Have in This Release 10
5. Requirement Classification 10
5.1 Functional Requirements 10
Signal Parameter Collection 10
Real-Time Signal Strength Display 11
Coverage Map Visualisation 11
Offline Data Storage and Synchronisation 12
Background Data Collection Service 12
Secure Data Transmission 12
Signal Quality History Log 13
Coverage Hole Detection and Notification 13
5.2 Non-Functional Requirements 13
Performance 13
Battery Efficiency 14
Data Consumption 14
Usability 14
Privacy and Security 14
Reliability 15
Compatibility 15

Scalability 15
6. Stakeholder Validation 15
End User Validation 16
Technical Stakeholder Validation 16
Mobile Network Operators 16
Backend and System Administrators 17
Academic Supervisor Alignment 18
7. Use Case Summary 18
8. Data Flow Overview 19
9. Challenges and Risks 20
10. Conclusion 20
11. References 21

1. Introduction

Requirement analysis is the systematic process of examining, refining, and formalising the raw requirements obtained during the requirement gathering phase. It bridges the gap between what stakeholders want and what developers can technically deliver. In the context of this project, the requirement analysis phase takes the data collected through surveys, brainstorming sessions, and reverse engineering of existing applications during Task 2 and transforms it into a structured, validated, and prioritised set of requirements that will directly guide the design and implementation of the proposed crowdsensed drive test mobile application.

The proposed application is designed to address a clearly identified gap in Cameroon's mobile network monitoring landscape. By enabling ordinary smartphone users to passively contribute geo-referenced signal measurements as they move about their daily environments, the application will provides a scalable, costeffective alternative to traditional drive testing. The requirement analysis phase ensures that every feature built into the application is justified, technically feasible, free of ambiguity, and aligned with the genuine needs of the stakeholders identified in Task 2.

This report documents the full requirement analysis for the crowdsensed drive test mobile application. It reviews and evaluates the requirements gathered, identifies inconsistencies and missing information, prioritises requirements by importance and feasibility, classifies requirements into functional and non-functional categories, develops the Software Requirement Specification (SRS), and validates the requirements with relevant stakeholders. The output of this phase forms the authoritative baseline for system design, user interface development, and backend architecture in the subsequent tasks.

2. Review and Analysis of Gathered Requirements

The requirements gathered during Task 2 were subjected to a thorough review against four quality criteria: completeness, clarity, technical feasibility, and dependency relationships. This review process is essential for identifying gaps, resolving ambiguities, and ensuring that the set of requirements is internally consistent and ready for design.

2.1 Completeness

A complete set of requirements captures all the behaviours, constraints, and qualities the system must possess without leaving gaps that would force the development phase to make uninformed

decisions. The review of Task 2 findings revealed that while the survey data provided strong coverage of user facing functional preferences such as real time signal display, coverage mapping, and background collection, several important system level requirements were not explicitly surfaced by respondents but are nonetheless essential to the operation of the application.

Specifically, the gathered requirements did not address user authentication and account management, data anonymisation protocols before transmission to the backend, the mechanism for handling API deprecation across different Android versions, or the process for managing firmware differences in how Android devices expose signal parameters. These gaps were identified through brainstorming sessions and reverse engineering of comparable applications such as opensignal, Gnet track, and have been incorporated into the complete requirements set documented in this report.

2.2 Clarity

Clarity requires that each requirement be stated precisely and without ambiguity so that developers, testers, and stakeholders all derive the same understanding from it. Several requirements as originally stated in Task 2 were found to be insufficiently specific. For instance, the requirement for “real time signal strength display” did not specify the refresh rate, the signal parameters to be displayed, the visual format, or the threshold values that should trigger visual alerts. Similarly, the requirement for “offline data storage” did not define the maximum local storage buffer size, the data retention policy when storage is full, or the priority rules for synchronisation when connectivity is restored.

Each of these ambiguous requirements has been refined in the functional and non-functional requirements sections of this report with explicit, measurable specifications that leave no room for interpretive error during implementation.

2.3 Technical Feasibility

Technical feasibility assessment ensures that each requirement can realistically be implemented within the constraints of the target platform, available APIs, team expertise, and project timeline. The most significant feasibility consideration for this project concerns the availability of signal measurement APIs across Android device versions and manufacturers.

Android exposes cellular signal parameters such as RSRP, RSRQ, and SINR through the TelephonyManager API, but the level of detail available varies significantly between Android versions. Devices running Android 9 and above expose significantly richer signal data through

the getAllCellInfo() and registerTelephonyCallback() methods. Devices below Android 9 provide limited signal data, particularly for 4G LTE measurements. Furthermore, some device manufacturers restrict access to certain parameters at the hardware driver level. These constraints have been incorporated into the requirements by defining minimum Android API level support and establishing fallback behaviour for devices with limited API access.

The coverage prediction requirement, which involves applying machine learning models to aggregated crowdsourced data, was also assessed for feasibility. Given the project timeline and the academic context, it was determined that a rule-based threshold classification approach is more feasible than a full machine learning pipeline for the initial implementation, with the architecture designed to accommodate more sophisticated prediction models in future iterations.

2.4 Dependency Relationships

Dependency analysis identifies which requirements must be implemented before others can function correctly. The following key dependencies were identified during the review process. The GPS geo-tagging requirement is a hard dependency for the coverage map visualisation and coverage prediction requirements, as neither can function without location-tagged measurements. The offline data storage requirement is a dependency for the background data collection requirement, since collected data cannot be discarded if no network connection is available. The user consent and permission management requirement is a dependency for all data collection requirements, as no data may be collected without explicit user authorisation under applicable data protection principles. The backend data transmission requirement depends on the data schema definition requirement, which in turn depends on the signal parameter collection requirement. These dependency relationships are visualised in the system context diagram below.

3. Inconsistencies, Ambiguities and Missing Information

The review process identified several inconsistencies and instances of missing information that required resolution before the requirements could be considered fit for use in system design.

It is necessary to evaluate the collected requirements for inconsistencies, ambiguities, and missing information. This stage is important because requirements gathered from surveys, interviews, and field experiments may contain unclear statements, conflicting expectations, or incomplete details. If these issues are not identified early, they can negatively affect system

design, implementation, and user acceptance. Unlike requirement prioritization, which focuses on ranking requirements according to importance, this stage focuses on improving the quality and clarity of the requirements themselves. The goal is to ensure that the development team has a complete, consistent, and understandable set of requirements before implementation begins.

Identifying Inconsistencies

Inconsistencies occur when two or more requirements conflict with each other or cannot be fully achieved at the same time. During the analysis of survey responses and experimental observations, several inconsistencies were identified.

One major inconsistency involved the balance between continuous sensing and battery efficiency. Some participants requested real time data collection to improve coverage prediction accuracy, while others complained that continuous sensing rapidly drained battery power. These two expectations conflict because higher sensing frequency increases energy consumption. To resolve this issue, adaptive sensing intervals were proposed so that the app could reduce sensing frequency when the device battery is low while still maintaining acceptable prediction accuracy.

Another inconsistency was observed between detailed location tracking and user privacy. Accurate coverage prediction requires precise GPS data, but many users expressed discomfort with constant location monitoring. This created a conflict between the need for accurate datasets and the requirement for user trust and privacy protection. The analysis therefore recommended anonymization and encryption techniques to reduce privacy risks while preserving useful network data.

In addition, some stakeholders from already existing system review emphasized advanced predictive analytics using machine learning, whereas others prioritized a lightweight application that could run efficiently on low end smartphones. Machine learning models often require additional processing power and storage, which may affect application performance. This inconsistency highlighted the need for optimization techniques and cloud-assisted processing.

Identifying Ambiguities

Ambiguities arise when requirements are vague, unclear, or open to multiple interpretations. Several ambiguous requirements were identified during analysis.

For example, survey participants frequently requested an “easy-to-use interface.” However, the term easy-to-use is subjective because different users may interpret usability differently depending on their technical experience. Some users may prefer minimal controls, while others may expect advanced customization features. To reduce ambiguity, the requirement was refined to specify measurable usability goals such as simple navigation, minimal setup steps, and readable interface elements.

Another ambiguous requirement concerned “accurate coverage prediction.” The word accurate was not clearly defined in the gathered requirements. Accuracy could refer to signal strength estimation, geographic precision, prediction consistency, or comparison with real network measurements. To clarify this requirement, measurable performance indicators such as prediction error rate and signal estimation reliability needed to be specified.

Similarly, the requirement for “low power consumption” lacked clear measurement standards. Without defining acceptable battery usage levels, developers may interpret the requirement differently. The analysis therefore suggested defining limits for background energy usage during sensing operations.

Identifying Missing Information

Missing information refers to important details that were not fully captured during requirement gathering. Identifying these gaps helps prevent implementation problems later in development.

One missing area was the absence of clear data retention policies. While privacy and security were prioritized, the requirements did not specify how long user data would be stored, who could access it, or when it would be deleted. These details are necessary to ensure compliance with data protection standards and improve user trust.

Another missing element involved network failure handling. The gathered requirements discussed offline functionality, but they did not clearly explain how data synchronization would occur once internet connectivity was restored. Additional clarification was needed regarding caching limits, synchronization timing, and duplicate data prevention.

The requirements also lacked detailed information about compatibility with different smartphone operating systems and hardware configurations. Since users may participate using devices with varying processing power, storage capacity, and Android versions, compatibility requirements should be explicitly defined.

Furthermore, there was limited information about how users would be motivated to continuously contribute sensing data. Although user participation is essential for crowd-sensing systems, the requirements did not address incentive mechanisms such as rewards, gamification, or feedback systems that encourage long-term participation.

Importance of Resolving These Issues

Resolving inconsistencies, ambiguities, and missing information improves the overall quality of the software requirements specification. Clear and consistent requirements reduce misunderstandings between developers, stakeholders, and users. They also help ensure that the final application meets user expectations while remaining technically feasible.

For the Crowdsensed Drive Test Mobile App, addressing these issues is especially important because the system depends heavily on user participation, accurate sensing, and trust in data handling practices. By refining unclear requirements and filling missing gaps, the project becomes more reliable, scalable, and easier to implement successfully.

4. Requirement Prioritisation

Requirements were prioritised using the MoSCoW method, which classifies requirements into four categories: Must Have, Should Have, Could Have, and Won't Have (in this release). This prioritisation reflects both the importance of each requirement to the core value proposition of the application and its feasibility within the current project timeline.

4.1 Must Have

These requirements are non-negotiable and must be implemented for the application to deliver its core value. They include collection of RSRP, RSRQ, SINR, RSSI, and Cell ID measurements from the device telephony API, GPS-based geo-location tagging of all measurements, local offline storage of measurements when internet is unavailable, encrypted transmission of data to the backend server; user consent and permission management for location and background process access, a real-time signal strength display on the application

home screen; and a basic coverage map showing collected data points colour-coded by signal strength.

4.2 Should Have

These requirements are important but not critical for the initial release. They include automatic background data collection as a foreground service; signal quality history logs accessible to the user; Wi-Fi-preferred data synchronisation to minimise mobile data consumption; battery optimisation through adaptive sampling intervals; data anonymisation before transmission; and network generation detection and labelling (2G, 3G, 4G, 5G).

4.3 Could Have

These requirements would enhance the application but will only be included if time and resources permit. They include user notifications when entering a detected coverage hole area; operator comparison dashboards showing signal quality by provider; bilingual interface support in English and French; export of personal signal history as a CSV file; and integration with social sharing features to encourage community participation.

4.4 Won’t Have in This Release

These requirements have been explicitly deferred to future versions. They include iOS application development; full machine learning-based coverage prediction using trained neural network models; integration with operator network management systems through formal APIs; and user reward or incentive mechanisms for data contribution.

5. Requirement Classification

Requirements have been classified into two primary categories: functional requirements, which describe what the system must do, and non-functional requirements, which describe how the system must perform its functions. This classification directly informs the system design and testing strategies in subsequent tasks.

5.1 Functional Requirements

Signal Parameter Collection

The application shall collect the following cellular signal parameters from the device Telephony Manager API at configurable intervals of between one and thirty seconds: Reference Signal Received Power (RSRP) in dBm, Reference Signal Received Quality

(RSRQ) in dB, Signal-to-Interference-plus-Noise Ratio (SINR) in dB, Received Signal Strength Indicator (RSSI) in dBm, Physical Cell Identifier (PCI), Tracking Area Code (TAC), Cell Identity (Cell ID), and network generation type (2G, 3G, 4G/LTE, 5G where available). Collection shall be supported on all Android devices running API level 26 or above, with graceful degradation for devices that do not expose all parameters.

GPS Geo-Location Tagging

Every signal measurement record collected by the application shall be tagged with the corresponding geographic coordinates (latitude and longitude accurate to at least five decimal places), altitude in metres above sea level, device heading in degrees, and a UTC timestamp accurate to the nearest second. Location data shall be obtained using the Android Location Services API with a minimum accuracy threshold of fifty metres. If location accuracy falls below this threshold, the measurement record shall be flagged as low-accuracy but retained. Data collection shall be suspended if location permission is denied or revoked by the user.

Real-Time Signal Strength Display

The application home screen shall display the current signal strength and quality metrics in real time with a maximum refresh interval of five seconds. The display shall include a graphical signal strength gauge using a colour-coded indicator

Colour	Strength	Signal interval
Green	Strong	Above -85dBm
Yellow	Moderate	-85 and -100dBm
Red	Weak	Below -100dBm

The current network generation type, the name of the connected operator, the Cell ID of the serving tower, and numeric values for RSRP, RSRQ, and SINR. The display shall update automatically when the device changes cells or network generation.

Coverage Map Visualisation

The application shall include a map screen that renders collected signal measurements as a colour-coded heat map overlay on a base map of the user's geographic area. Signal strength categories shall be visually distinguished: areas with RSRP above -85 dBm shall be displayed

in green, areas between -85 and -100 dBm in yellow, and areas below -100 dBm in red. The map shall support standard pan and zoom interactions. Data points shall be rendered as semi-transparent circles with a radius proportional to the estimated coverage area of the measurement. Users shall be able to filter the map view by date range, network operator, and network generation type.

Offline Data Storage and Synchronisation

The application shall implement a local SQLite database to store measurement records when an internet connection is unavailable. The local buffer shall accommodate a minimum of one hundred measurement records before triggering a storage management alert. When internet connectivity is restored, the application shall automatically initiate a background synchronisation process that transmits all locally buffered records to the backend server in batches of five hundred records. Successfully transmitted records shall be removed from the local buffer after server acknowledgement. In the event of synchronisation failure, the application shall implement an exponential backoff retry strategy with a maximum retry interval of thirty minutes.

Background Data Collection Service

The application shall implement data collection as an Android foreground service that displays a persistent status notification indicating that monitoring is active. The foreground service shall survive device screen-off states and shall be configured to restart automatically after device reboots using a broadcast receiver. Users shall be able to pause and resume the collection service from both the application interface and the persistent notification. The service shall implement Android's WorkManager API for scheduled synchronisation tasks to comply with Android background execution limits.

Secure Data Transmission

All data transmitted between the mobile client and the backend server shall be encrypted using TLS 1.2 or above over HTTPS. The application shall authenticate each transmission using a device-specific token issued at first registration. The token shall be stored securely in the Android Keystore system and shall not be stored in shared preferences or any unprotected storage location. The transmission payload shall use JSON format with field-level validation on both the client and server sides. The backend server shall return an acknowledgement response for each successfully received batch, and the client shall log transmission failures for diagnostic purposes.

User Consent and Permission Management

During the first application launch, users shall be presented with a clear and accessible onboarding screen that explains in plain language what data is collected, how it is processed, where it is transmitted, and who may access it. Users must explicitly grant the following permissions before data collection begins: ACCESS_FINE_LOCATION, ACCESS_BACKGROUND_LOCATION,READ_PHONE_STATE, FOREGROUND_SERVICE, and INTERNET. Each permission request shall be accompanied by a brief explanation of why it is needed. Users may withdraw consent at any time through the application settings screen, at which point all data collection shall immediately cease and locally buffered data shall be deleted upon user confirmation.

Signal Quality History Log

The application shall maintain a locally accessible history log of the user's signal quality measurements over time. The history screen shall display a time-series graph showing RSRP values over the past twenty-four hours by default, with options to extend the view to seven days or thirty days. Users shall be able to tap on any point in the graph to see the full measurement record including all signal parameters, GPS coordinates, operator, and network generation at that time. The history log shall be stored in the local SQLite database and retained for a maximum of ninety days on the device.

Coverage Hole Detection and Notification

The backend server shall implement a coverage hole detection algorithm that analyses the aggregated dataset to identify geographic areas where the median RSRP value across all measurements falls below -110 dBm and at least three independent device measurements have been recorded within a radius of two hundred metres. Detected coverage holes shall be stored in the backend database with their centroid coordinates, estimated radius, first detection date, and contributing measurement count. The mobile application shall download the list of detected coverage holes at application startup and shall issue a local push notification when the user enters a coverage hole zone as determined by their current GPS position.

5.2 Non-Functional Requirements

Performance

The application home screen shall render with a cold start time of no more than three seconds on a mid-range Android device with 2GB of RAM. Signal parameter collection shall introduce

no more than two percent additional battery drain per hour compared to the device's idle baseline when the collection interval is set to ten seconds. The coverage map screen shall render up to ten thousand data points without frame rate drops below thirty frames per second. Data synchronisation batches of five hundred records shall complete transmission within ten seconds on a standard 4G connection.

Battery Efficiency

The application shall implement adaptive sampling intervals that automatically increase the collection interval from the default ten seconds to thirty seconds when the device has been stationary for more than five minutes, as determined by the ActivityRecognitionAPI. When the device battery level falls below twenty percent, the application shall automatically switch to a power-saving mode that reduces collection frequency to sixty-second intervals and disables the coverage map rendering engine. Users shall be notified when power-saving mode is activated.

Data Consumption

The application shall consume no more than five megabytes of mobile data per day during normal operation with the default ten-second collection interval. Synchronisation shall be preferentially scheduled over Wi-Fi connections. Users shall be able to set a daily mobile data consumption limit for synchronisation in the application settings, with a default limit of two megabytes per day. The application shall display the cumulative data transmitted in the current calendar month on the settings screen.

Usability

The application shall be operable by a non-technical user without any prior training beyond the onboarding flow. All primary functions shall be accessible within two taps from the home screen. Text displayed within the application shall use a minimum font size of fourteen points to ensure readability on small screens. The application shall support both portrait and landscape orientations on all screens. The installed application size shall not exceed fifty megabytes. The application interface shall be available in English, with French language support as a future enhancement.

Privacy and Security

No personally identifiable information shall be transmitted to the backend server. Device identifiers shall be pseudonymised using a one-way cryptographic hash before transmission, ensuring that individual users cannot be identified from the server-side dataset. The local

SQLite database shall not be accessible to other applications installed on the device. All sensitive configuration values including API endpoints and authentication tokens shall be stored in the Android Keystore system and obfuscated in the compiled application binary using ProGuard or R8. The application shall implement certificate pinning to prevent man-in-the-middle attacks.

Reliability

The application shall be capable of operating continuously for a minimum of twenty-four hours without crashing or requiring a restart on any device meeting the minimum hardware specification. The application shall recover automatically from transient errors including temporary network unavailability, GPS signal loss, and telephony service interruptions without requiring user intervention. Critical errors that cannot be recovered automatically shall be logged to a local diagnostic file and the user shall be presented with a clear error message and recommended resolution steps.

Compatibility

The application shall support Android versions 8.0 (API level 26) through the latest stable Android release at the time of deployment. The application shall function correctly on both single-SIM and dual-SIM devices. The user interface shall be responsive and correctly rendered on screen sizes ranging from 4.7 inches to 6.9 inches with screen densities from 160 dpi to 640 dpi. The application shall not require any third-party applications or services beyond those available through the Google Play Store.

Scalability

The backend server architecture shall be designed to accommodate growth from an initial user base of one hundred devices to a maximum of one hundred thousand concurrent contributing devices without architectural redesign. The database schema shall support partitioning by geographic region and time period to maintain query performance as the dataset grows. The API design shall be versioned to allow future modifications without breaking compatibility with older application versions.

6. Stakeholder Validation

Requirement validation is the process of confirming with relevant stakeholders that the documented requirements accurately and completely represent their needs and expectations before development begins. For this academic project, stakeholder validation was conducted

through a structured peer review process involving the project team, a review of the Task 2 survey findings for alignment, and a walkthrough of the requirements with the academic supervisor.

End User Validation

End user validation was performed by cross-referencing the functional requirements with the survey findings from Task 2. Each Must Have and Should Have functional requirement was traced back to at least one survey finding to confirm that it is grounded in genuine user need rather than assumed necessity. The real-time signal display requirement is validated by the 76.2 percent survey response identifying it as the most desired feature. The coverage map requirement is validated by the 35.5 percent of respondents who selected it as a desired feature. The offline storage requirement is validated by the geographic distribution of reported coverage problems, which confirms that users routinely operate in areas without internet connectivity. The battery efficiency requirement is validated by user concerns about battery drain raised during the survey, which were among the most frequently cited barriers to adoption.

Technical Stakeholder Validation

The technical requirements relating to signal parameter collection, API level compatibility, data schema, and backend architecture were reviewed by the project team against Android developer documentation and the technical specifications of comparable applications studied during the reverse engineering exercise in Task 2. The minimum Android API level of 26 was validated against the Android version distribution data for Cameroon, which indicates that the majority of active Android devices in the country run Android 8.0 or above. The TLS 1.2 encryption requirement was validated against current Android security best practices and the requirements of standard data protection frameworks.

Mobile Network Operators

Mobile network operators such as MTN Cameroon and Orange Cameroon are considered indirect but highly important stakeholders in the proposed crowdsensed drive test system. During the stakeholder validation process, the requirements associated with network operators were carefully reviewed and analysed in relation to the objectives of the application. It was confirmed that these operators can benefit significantly from the crowdsourced network measurements generated by the system because the collected data can help identify weak coverage zones, detect network inconsistencies, optimise antenna placement, and improve overall service delivery in underserved areas. The validation process further confirmed that the

application aligns with existing industry practices used in network performance monitoring and quality-of-service assessment. The signal parameters selected for collection, including RSSI, RSRP, RSRQ, SINR, Cell ID, and network type, were validated as relevant and useful indicators for network coverage analysis. The operators' technical standards and existing approaches to network measurement therefore influenced the structure of the backend database schema and analytics framework adopted for the project. Stakeholder validation also confirmed that the crowdsensing approach offers a more scalable and cost-effective alternative to traditional drive testing methods, particularly for large geographic regions and rural environments where conventional testing may be difficult or expensive to perform.

Backend and System Administrators

The backend developers and system administrators responsible for managing the server infrastructure, database systems, APIs, and analytics services represent another important stakeholder group. During stakeholder validation, the technical requirements related to backend management, scalability, security, and system reliability were carefully examined to ensure that the proposed infrastructure can support large-scale crowdsensed data collection and processing.

The validation process confirmed that the backend architecture must support secure communication between the mobile application and cloud servers through encrypted APIs and authentication mechanisms. It was also validated that the system requires reliable database management structures capable of handling large volumes of georeferenced signal measurements generated by multiple users simultaneously.

Stakeholder discussions further confirmed the need for efficient synchronization mechanisms, fault tolerance, automated backups, and server monitoring tools in order to ensure consistent system performance and reliability. The administrators also validated the importance of designing the backend infrastructure in a modular and scalable manner so that future features such as advanced analytics, AI-based prediction models, and expanded geographic datasets can be integrated easily without major architectural redesign.

The validation of these requirements directly influenced several non-functional requirements of the system, including maintainability, scalability, performance optimization, and data security.

Regulatory Bodies

The Agence de Régulation des Télécommunications (ART) represents an important regulatory stakeholder because the proposed system involves the collection, processing, and transmission of georeferenced mobile network data. During stakeholder validation, regulatory considerations relating to data privacy, telecommunications compliance, and user consent management were carefully analysed in order to ensure that the proposed application operates within acceptable legal and ethical boundaries.

The validation process confirmed that the application must implement transparent privacy policies clearly explaining how user data is collected, processed, stored, and protected. It was also validated that users must explicitly grant permissions before any location or signal measurements are collected from their devices. These requirements significantly influenced the system's privacy architecture and consent management mechanisms.

Furthermore, the stakeholder validation process confirmed the need for secure data encryption during transmission and storage in order to reduce risks associated with unauthorized access or misuse of collected information. Compliance with telecommunications regulations and responsible handling of crowdsensed data were therefore validated as essential requirements for the successful deployment and acceptance of the system within Cameroon.

Academic Supervisor Alignment

The complete requirements set documented in this report is aligned with the project brief as defined in the course outline for CEF440 and the project topic assigned by Dr. Valery Nkemeni. The scope of functional requirements covers the full lifecycle of the crowdsensed data collection and processing pipeline as described in the project title, from smartphone-based measurement collection through backend transmission to coverage prediction output. The non-functional requirements address the quality attributes most relevant to a mobile application operating in the Cameroonian network environment, including battery efficiency, low data consumption, and offline capability.

7. Use Case Summary

Based on the validated requirements, the following primary use cases have been identified for the proposed application. These use cases will be elaborated into full UML Use Case Diagrams in Task 4.

- **Install and Onboard:** A new user installs the application, completes the onboarding consent flow, grants required permissions, and begins contributing signal data.
- **Monitor Signal in Real Time:** A user opens the application and views their current signal strength, network type, operator, and serving cell information on the home screen.
- **View Coverage Map:** A user navigates to the map screen to view the colour-coded signal coverage map of their area and identifies nearby coverage holes.
- Collect Data in Background: The application runs as a foreground service and passively collects geo-referenced signal measurements while the user goes about their daily activities.
- **Synchronise Data with Backend:** The application detects internet connectivity and automatically uploads locally buffered measurements to the backend server.
- **View Signal History:** A user accesses their personal signal quality history log and reviews trends over the past day, week, or month.
- Receive Coverage Hole Notification: The application detects that the user has entered a known coverage hole area and issues a push notification to inform them.
- Manage Privacy Settings: A user accesses the settings screen to review data collection status, adjust the sampling interval, set data consumption limits, or withdraw consent and stop collection.

8. Data Flow Overview

The data flow through the proposed system can be summarised in four stages. In the first stage, the Android device collects signal parameters from the telephony hardware via the TelephonyManager API and combines them with GPS coordinates from the Location Services API to produce a geo-referenced measurement record. In the second stage, the measurement record is written to the local SQLite database with a pending synchronisation flag. In the third stage, when internet connectivity is available, the application retrieves pending records from the local database, packages them into a JSON batch payload, and transmits them to the backend REST API over HTTPS. In the fourth stage, the backend server validates, stores, and processes the received records, updating the coverage map and running the coverage hole detection algorithm on the new data. The processed results are made available through the backend API for display in the mobile application's map and statistics screens.

9. Challenges and Risks

Several challenges and risks have been identified during the requirement analysis phase that may affect the implementation and deployment of the proposed application.

The most significant technical risk concerns API fragmentation across Android devices. The availability and accuracy of signal parameters such as RSRP and SINR vary considerably across device manufacturers and Android versions, which may result in inconsistent data quality across the contributing device population. This risk is mitigated by implementing robust data validation on both the client and server sides, flagging measurements from devices with known API limitations, and designing the coverage prediction algorithm to be tolerant of sparse or incomplete data.

A second risk concerns user retention and sustained data contribution. The application's value proposition depends on a sufficiently large and geographically distributed user base. If adoption is concentrated among a small group of users in a limited area, the coverage map will lack the density needed for meaningful prediction. This risk is addressed in the requirements by designing the onboarding experience to clearly communicate the collective benefit of participation and by including the coverage hole notification feature as an immediate personal incentive for continued use.

A third risk relates to data privacy perception. Given that the application collects location data continuously, some users may be reluctant to maintain long-term participation despite initial consent. This risk is mitigated by implementing data anonymisation at the point of collection, providing transparent data usage reporting in the settings screen, and giving users full control over collection frequency and the ability to delete their locally stored data at any time.

10. Conclusion

The requirement analysis phase of this project has produced a comprehensive, validated, and prioritised set of software requirements that will serve as the authoritative foundation for all subsequent development phases. Through systematic review of the requirements gathered in Task 2, this report has resolved ambiguities, filled identified gaps, established dependency relationships, and classified requirements into a formal Software Requirement Specification aligned with the functional scope and quality expectations of the project.

The ten functional requirements and eight non-functional requirements documented in this report collectively define a mobile application that is technically feasible within the Android

platform constraints, responsive to the genuine needs of the target user population as evidenced by the 62-respondent survey, and capable of delivering meaningful value to mobile network operators and end users in Cameroon. The use cases, data flow overview, and prioritisation framework established in this report will directly inform the system modelling and design activities to be carried out in Task 4, where context diagrams, data flow diagrams, use case diagrams, sequence diagrams, class diagrams, and deployment diagrams will be developed to fully specify the architecture of the proposed system.

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